FEDERAL SPECIFICATION

TUBE, HYPODERMIC NEEDLE PROTECTING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers glass hypodermic needle protecting tubes for use in the sterilization of hypodermic needles.

1.2 Classification.

1.2.1 *Types, styles, and sizes.* — Tubes shall be of the following types, styles, and sizes, as specified (see 6.1) :

Type I. — For single pointed needles.

Style A.—Beaded edge.

Size 1.—3-inch.

Size 2.—4 1 1/16-inch.

Style B.—Plain edge.

Size 1.—4-inch.

Size 2.—5½-inch.

Type II.—For double pointed needles.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

DD-G-541 — Glass (Laboratory).

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Box, Fiberboard.

PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102 — Preservation, Packaging and Packing Levels.

Fed. Std. No. 123 — Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 — Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

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MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials: Publication:

ASTM Designation C149–50—Standard Method of Thermal Shock Test on Glass Containers.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material.—The tubes shall be fabricated of soda-lime glass in accordance with type II of Federal Specification DD–G–541 and shall be free from marked irregularities, seeds, striae, or bubbles.

3.2 Style, design, and dimensions.—Tube shall be a cylindrical glass tube, with constriction in diameter, suitable for sterilization of a hypodermic needle. The constriction shall act as a resting point for the hub of the needle in such a manner that the point of the needle shall not rest on the bottom of the tube.

3.2.1 *Type I.* — Type I tubes shall be suitable for use with single pointed needles.

3.2.1.1 *Style A.* — Style A tubes shall be constructed as shown in figure 1, with a beaded edge at the open end. Size 1 tubes shall be suitable for containing a hypodermic needle with a $1\frac{1}{2} \pm \frac{1}{16}$ inch cannula, and size 2 tubes shall be suitable for containing a hypodermic needle with a $3\frac{1}{4} \pm \frac{1}{8}$ inch cannula.

3.2.1.2 *Style B.* — Style B tubes shall be constructed as shown in figure 2. Size 1 tubes shall be suitable for containing a hypodermic needle with a $2\frac{1}{4} \pm \frac{5}{64}$ inch cannula, and size 2 tubes shall be suitable for containing a hypodermic needle with a $3\frac{1}{2} \pm \frac{1}{8}$ inch cannula.

3.2.2 *Type II.* —Type II tubes shall be suitable for use with a 20 gage, $1 \pm \frac{1}{32}$ inch cannula and $1\frac{1}{2} \pm \frac{1}{32}$ inch cannula double pointed, blood-collecting needle used with serological blood collecting sets. Type II tubes shall be constructed as shown in figure 3.

3.3 Rubber stopper.—When required (see 6.1), a rubber stopper capable of making an airtight cover and capable of maintaining the sterility of the needle shall be supplied with each tube.

3.4 Resistance to thermal shock. — The tubes shall not break (due to temperature changes) when submitted to the thermal shock test, as specified in 4.4.1.

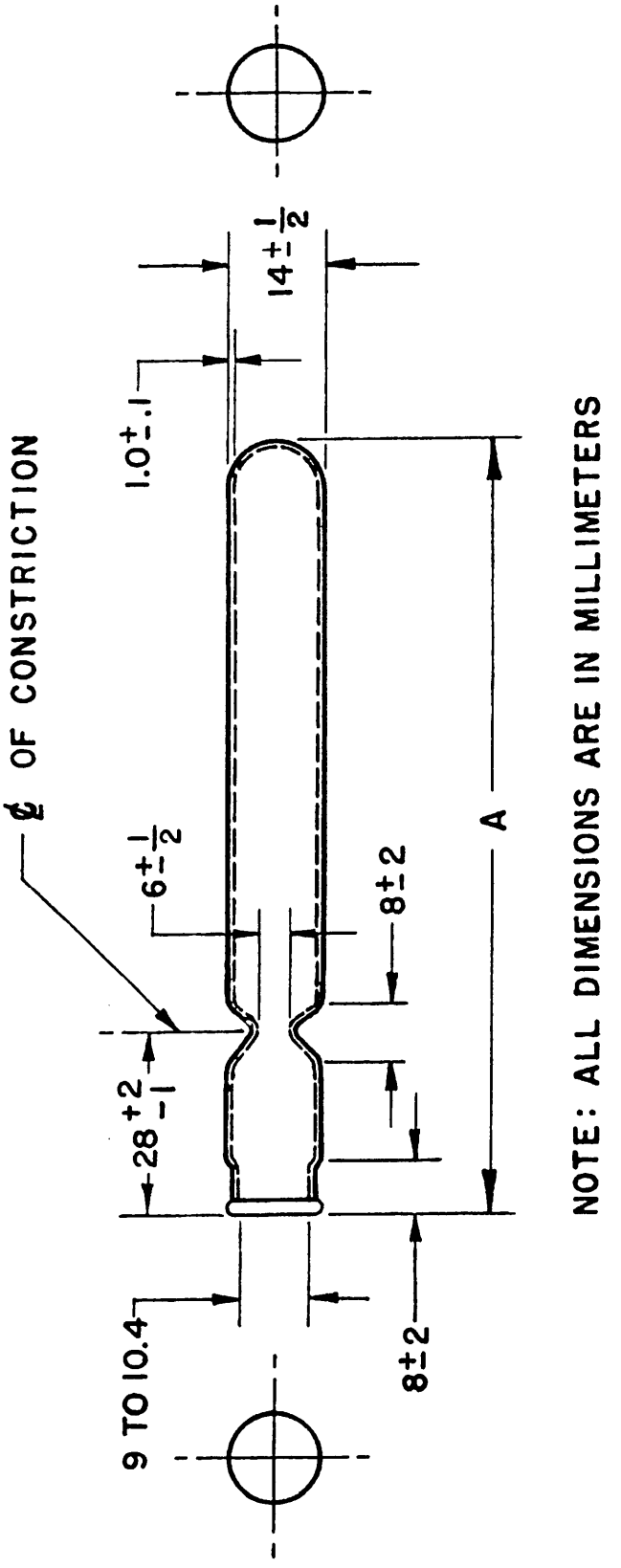
3.5 Annealing.—The tubes shall be examined for strain as specified in 4.4.2. A moderate amount of evenly distributed longitudinal strain is acceptable. Any ring strain or severe longitudinal strain shall not be acceptable.

3.6 Workmanship.—Workmanship shall be first class throughout. Tubes shall be free from defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection.—The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records

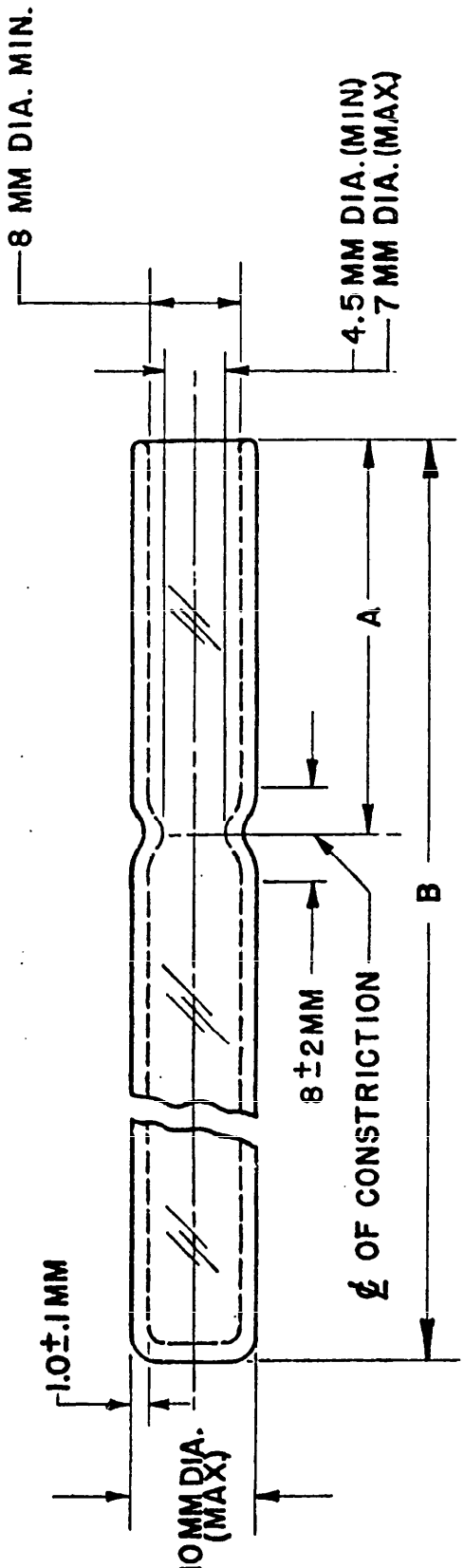
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SIZE	DIM. A
1	85±5
2	122±5

FIGURE 1. — Type I, style A.

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SIZE	DIM. A	DIM. B
1	33±2MM	100±3MM
2	40±2MM	141±3MM

FIGURE 2. — Type I, style B.

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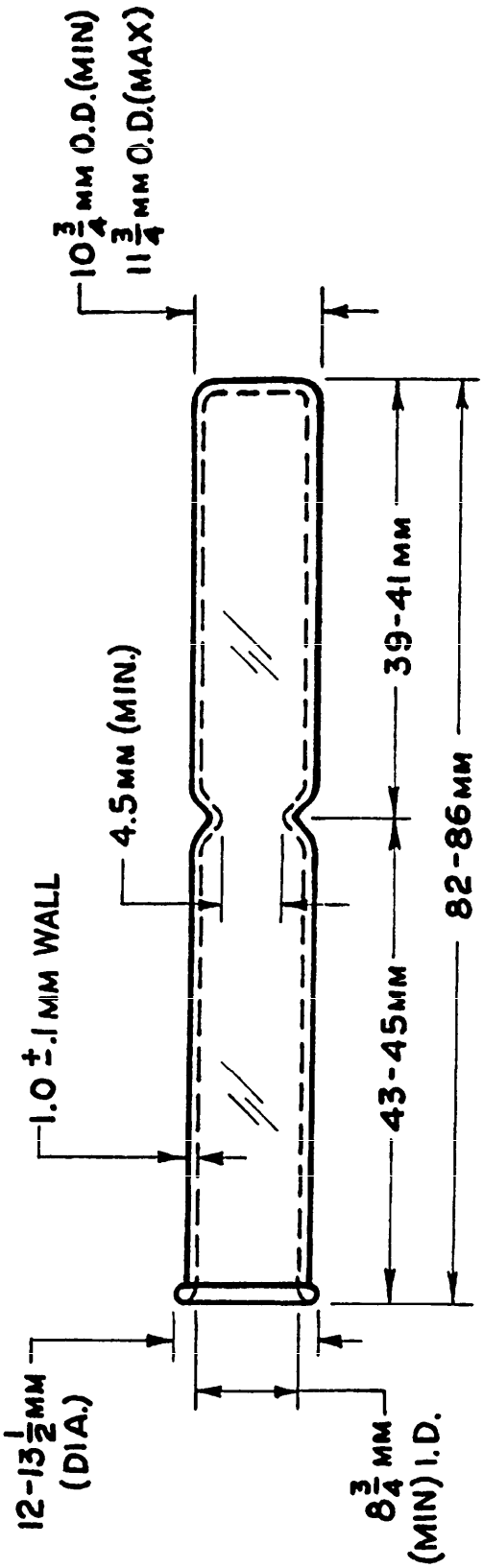


FIGURE 3. — Type II.

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of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.**—Unless otherwise specified, the unit of product for inspection purposes shall be the individual tube.

4.2.1 *For examination.* — Sampling for examination shall be conducted in accordance with Military Standard MIL–STD–105.

	Inspection level	AQL (percent defective)
For visual examination:		
Major A defects	II	1.0
Major B defects	L1	1.0
Minor A defects	II	2.5
Minor B defects	L1	2.5
For dimensional examination	L2	2.5

4.2.2 *For tests.* — Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105. Requirement applicable to the individual unit only.

Component	Characteristic	Req.	Test procedure	Unit of product	Insp. level	AQL
Tube	Thermal shock	3.4	4.4.1	1-tube	L3*	**
Tube	Annealing	3.5	4.4.2	1-tube	L3*	**

* A minimum of three units shall be tested regardless of lot size.
** Acceptance number shall be zero for all sample sizes.

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4.2.3 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 **Examination.**—The tubes shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 *Classification of defects.* — Examination shall be conducted in accordance with the following classifications of defects:

*Categories and defects****

Major A: Unit of product shall be one tube.

101. Tube not free of localized thin walled areas, cracks, breaks, or chipped sections, and sharp edges.
102. Tube not free of badly deformed sections or areas.
103. Point of hypodermic needle (of cannula length corresponding to tube size) fails to clear the bottom when needle is inserted into tube.
104. Where applicable, rubber stopper fails to fit securely onto tube.

Major B: Unit of product shall be one intermediate package and contents.

151. Marking on intermediate and unit packages not correct.

Minor A: Unit of product shall be one tube.

201. Tube not free from distortion, excessive amounts of seeds, striae, bubbles, and tears.

Minor B: Unit of product shall be one intermediate package and contents.

251. Label or printing not legible and free of defacing marks, scratches, and tears.

*** Examination is not restricted to the classified possible defects listed above.

4.3.2 *Dimensional examination.* — The tubes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests.—Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Thermal shock test.* — Tubes shall be tested in accordance with American Society For Testing Materials (ASTM) Standard Method of Thermal Shock Test on Glass Containers, ASTM Test Designation C149–50. The hot bath shall be at a temperature of 145° Fahrenheit.

4.4.2 *Annealing test.* — Tubes shall be examined for annealing strains with a polariscope.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment).

5.1 Packaging.

5.1.1 *Unit of issue.* — Twelve tubes, as specified, constitute one unit of issue.

5.1.2 *Packaging quantities.* — The number of units of issue indicated in table I, shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

TABLE I.—*Packaging quantities*

Type	Style	Size	Unit package	Intermediate package	Exterior container
I	A or B	1	1 unit	36 units	144 units
I	A or B	2	1 unit	36 units	144 units
II	--	--	1 unit	48 units	96 units

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5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.* — Each unit (12 tubes) shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566, PPP–B–636, or PPP–B–676. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 *Intermediate package.* — Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 *Level C.* — Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 **Packing.**

5.2.1 *Level A.*

5.2.1.1 *Exterior container.* — Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.* — Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.* — Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table II, or PPP–B–636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking—nonmilitary.**

5.3.1 In addition to any special marking specified in the contract or order unit packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.4 **Marking—Military.**

5.4.1 *Unit package.* — Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

5.4.2 *Intermediate package.* — Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.4.3 *Exterior container.* — Exterior container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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(*a*) Title, number and date of this specification.

(*b*) Types, styles, and sizes required (see 1.2).

(*c*) Rubber stopper when required (see 3.3).

(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Transportation description. — Transportation description applicable to this item is:

Glassware, chemical laboratory.

Carload minimum weight 18,000 pounds, subject to Rule 34 of the Uniform Freight Classification.

Motor volume minimum weight 18,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIAN:
Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
Air Force.

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